Short Term Interest Rate Innovations Across Monetary Regimes

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Status: Documented reconstruction. Every estimate reported here is produced from source-compatible reconstructions rather than from the original article's own input files, which are not distributed. No result carries an exact-replication label, and the reasons are set out in Section 4.

Abstract

This paper asks whether short-term interest-rate innovations help explain the cross section of stock-market anomaly returns, and whether that relation is stable after the publication sample and across monetary regimes. Because Maio and Santa-Clara (2017) identify their data providers rather than their files, we do not attempt a numerical replication. We build the closest *documented reconstruction* obtainable from identifiable public inputs, hold every threshold and decision rule fixed in advance, and report each registered gate whether it passes or fails. On the 1972–2013 baseline the reconstruction recovers the published pattern qualitatively: the short-rate innovation carries a negative price of risk that is nominally significant under the article's Shanken correction, and it cuts cross-sectional pricing errors against the market model by roughly a half. Recovery of the published cells at printed precision is much weaker, particularly for pricing-error and fit statistics, so the exercise supports statements about patterns and not about arithmetic. Implementing the article's own useless-factor bootstrap sharpens that distinction rather than softening it: the reconstructed empirical p-values agree with all 45 published risk-price significance verdicts while matching far fewer printed digits, because a five-thousand-replication p-value carries more Monte Carlo noise than three printed decimals can resolve. What reconstructs is the inference, not the arithmetic. That pattern proves markedly less robust once economic materiality, alternative factor models, later data, beta-estimation error, and regime dependence are considered. The incremental-pricing standard registered in advance is not met on the headline asset set, and every registered traded multi-factor comparator attains a lower in-sample cross-sectional RMSE on the same seventy portfolios, albeit on richer factor sets. The post-2013 point estimates fail the registered compatibility gates, reversing sign for five of seven anomaly families. That failure is not explained by the measured data-vintage contribution, and the registered exposure-dispersion gate continues to pass; substantial beta-estimation uncertainty nevertheless prevents reading it as proof that the underlying pricing relation changed, and four of the five reversals are of signs the baseline never established. Rate exposures are regime dependent under a pooled interaction test, while regime-specific differences in the price of risk cannot be identified at feasible regime lengths. A registered out-of-sample falsification, refitting annually from a frozen 1999-12 training endpoint, returns a cross-sectional out-of-sample R^2 of 0.0039 against a historical-mean benchmark: the model beats a zero-return rule comfortably and the historical mean by a margin too small to be economically interesting. A calibrated simulation points to what these share: estimated rate betas have a reliability ratio of 0.386 even in the full sample, so most of their cross-sectional dispersion is estimation error, and the nominal Shanken interval stated coverage at no simulated sample size.

1 Introduction

Short-term interest rates are plausible state variables in intertemporal asset-pricing models because they summarize investment opportunities that matter for both discount rates and hedging demands. If portfolios formed on accounting or return-based anomalies have different exposures to those state variables, a rate innovation may help organize average excess returns beyond the market factor. The economic question is not only whether such a factor prices anomaly returns in one sample, but whether the relation survives changes in monetary regimes and source definitions.

Maio and Santa-Clara (2017) provide the starting point. Their article studies short-term interest-rate innovations and stock-market anomalies, and this paper uses that design as the baseline object to be reproduced before any extension is interpreted. The strict baseline window is January 1972 through December 2013. The extension window begins in January 2014 and is evaluated separately from the replication verdict.

The paper has three aims. First, it rebuilds the original baseline in a way that distinguishes exact replication from documented reconstruction. Second, it asks whether the short-rate factor is stable after the publication sample and across monetary regimes. Third, it evaluates whether any short-rate pricing evidence is robust to weak-factor diagnostics and comparator-model discipline. The high-frequency policy-information decomposition has been withdrawn from the design, because the only source that separates the components begins in 1990 and cannot reach the period supplying most of the variation being priced.

Four findings follow. On the baseline window the reconstruction recovers the published qualitative result: the short-rate innovation earns a negative price of risk of -0.6985 per month with a nominal Shanken t of -2.86, and it roughly halves cross-sectional pricing errors relative to the market model. Against that, the incremental-pricing standard we registered in advance is not met on the joint asset set, and every traded multi-factor comparator we examine attains a lower in-sample cross-sectional RMSE on the same portfolios, though on factor sets of higher dimension, so that comparison ranks fit rather than models. The post-2013 estimates then fail the registered compatibility gates. The measured data-vintage contribution does not explain that failure and the registered exposure-dispersion gate keeps passing, but the extension is 144 months long and its rate price carries a nominal Shanken t of -1.49, so we stop short of concluding that the underlying relation changed. Finally, short-rate exposures differ across monetary regimes, though the sample lengths available inside regimes are too short for the registered equivalence standard to resolve the size of the difference.

Two of these findings are negative, and we regard that as the substantive contribution rather than as a failed replication. The baseline qualitative pattern is recovered in the documented reconstruction; what does not survive is its extension. Every threshold, comparator, and decision rule was fixed before the corresponding estimate existed, so the negative findings are not the product of specification search, and we report each registered gate whether it passed or failed.

That ordering is checkable rather than asserted. The hypothesis registry, the economic thresholds, and the inference and bootstrap contracts were committed to a public repository on 2026-07-30 and 2026-07-31 and tagged as a release on 2026-07-31; the first cross-sectional estimate was produced on 2026-08-04. The commit history and that tag provide the timestamps, so a reader need not take the ordering on trust. We nonetheless describe the design as pre-specified in a versioned research workflow rather than as preregistered in the conventional sense, since it was not lodged with an external registry. The one threshold altered after estimation, the standalone-second-pass floor, is disclosed as such in the design-correction changelog and changes no registered classification.

2 Related literature

The first literature stream motivates the state variable. The ICAPM shows why assets can earn premia for covariance with variables that forecast future investment opportunities (Merton, 1973). Interest-rate applications of this idea are directly relevant because Lioui and Maio (2014) model interest-rate risk as a priced source of cross-sectional equity returns, and Maio and Santa-Clara (2017) apply short-rate innovations to anomaly portfolios.

The second stream concerns anomaly pricing and model comparison. The baseline article compares a short-rate ICAPM with standard factor models, so the extension must be evaluated against asset-pricing benchmarks rather than against the CAPM alone. Fama and French (2015) add profitability and investment factors to the Fama-French framework, while Hou et al. (2015) propose an investment approach that summarizes many anomalies with market, size, investment, and profitability factors. These benchmarks make the extension question sharper: does the short-rate factor add robust pricing content once comparator models, asset intersections, and portfolio definitions are held fixed?

The third stream concerns inference. Two-pass cross-sectional tests are exposed to generated-regressor and weak-factor problems. Fama and MacBeth (1973) provide the classic cross-sectional testing framework, Shanken (1992) derives corrections for beta-estimation error, and Kan and Zhang (1999) show that useless factors can appear priced in two-pass tests. Lewellen et al. (2010) argue that high cross-sectional fit and small pricing errors can be misleading when test assets share a strong characteristic structure. This paper therefore treats standardized exposure dispersion, rank, spanning, and influence diagnostics as conditions for interpretation rather than optional robustness checks.

The fourth stream concerns monetary regimes and high-frequency identification. Bernanke and Kuttner (2005) measure equity-market responses to unanticipated Federal Reserve policy actions, while Jarociński and Karadi (2020) separate policy and central-bank-information components using high-frequency announcement comovement. Swanson and Williams (2014) show why lower-bound episodes can change the informativeness of short-term rates for the broader term structure. Those papers motivate separating the aggregate AR rate innovation from identified announcement components and from regime-specific pricing evidence. Structural-change methods such as Bai and Perron (1998) provide a way to distinguish predeclared regime boundaries from estimated breaks.

The final stream is data vintage discipline. Official factor libraries publish current data, dated historical archives, and selected change notes rather than a complete revision history, so the retrieval vintage of a source has to be recorded before an estimate is labelled an exact replication or entered into a post-publication comparison (French, 2026). This paper therefore adds value beyond the original article by freezing a table-level replication target before asking whether the short-rate evidence generalizes over time and across regimes.

3 Original framework and economic mechanism

The original framework is a discrete-time ICAPM application in which the market excess return is paired with an innovation in a short-term interest-rate state variable. The article verifies two state variables: the federal funds rate and the three-month Treasury-bill rate. It motivates both series as proxies for investment opportunities and constructs innovations from first-order autoregressions.

For the federal-funds version, the article reports the state-variable law of motion as

$$FFR_{t+1} = 0.000 + 0.991 FFR_t + u_{t+1}^{FFR}, \quad R^2 = 0.98,$$

which this manuscript indexes for the return month as

$$u_t^{FFR} = FFR_t - \hat{a} - \hat{\rho} FFR_{t-1}.$$

This convention fixes the information timing: the return regression for month t uses the innovation from the same realized month and never relies on an implicit shift. The article reports the analogous Treasury-bill equation as

$$TB_{t+1} = 0.000 + 0.992 TB_t + u_{t+1}^{TB}, \quad R^2 = 0.98.$$

For test asset i, excess return $R_{i,t}^e$, market factor $R_{M,t}$, and federal-funds innovation u_t^{FFR}, the verified first-pass time-series regression is

$$R_{i,t}^e = \delta_i + \beta_{i,M} R_{M,t} + \beta_{i,FFR} u_t^{FFR} + \varepsilon_{i,t}.$$

The benchmark second-pass regression is a single OLS cross-sectional regression of full-sample average excess portfolio returns on full-sample first-pass betas, without an intercept:

$$\bar{R}_i^e = \hat{\beta}_{i,M} \lambda_M + \hat{\beta}_{i,FFR} \lambda_{FFR} + \alpha_i^{CS}.$$

Here δ_i is the time-series intercept, α_i^{CS} is the cross-sectional pricing error, and λ_{FFR} is the federal-funds innovation risk price. This benchmark is not described as repeated monthly Fama–MacBeth estimation unless a separate supplement target or robustness artifact explicitly implements repeated cross sections. The no-intercept restriction is part of the article's benchmark economic test, not a default regression convention.

The economic mechanism is an ICAPM-style hedging argument. A rate innovation can matter if it captures news about future investment opportunities and if anomaly portfolios differ in their exposure to that news. That statement is weaker than a monetary-policy interpretation. An autoregressive residual may combine policy news, information about the macroeconomy, data revisions, and persistence in the short-rate series. The paper therefore treats the baseline factor as a rate innovation until a separate high-frequency design justifies narrower language.

The article's primary test assets are value-weighted decile portfolios sorted on book-to-market, equity duration, earnings-to-price, long-term reversal, investment-to-assets, property-plant-and-equipment investment, and inventory growth. It also studies the joint system of all seven anomaly families and uses equal-weighted portfolios and additional appendix designs as robustness checks. The benchmark inference reports Shanken-corrected risk-price statistics, bootstrap p-values, a joint pricing-error statistic, and cross-sectional fit metrics.

The mechanism is not one channel. Short-rate innovations may proxy for discount-rate news, cash-flow news, financing-cost changes, working-capital conditions, monetary-policy information, or intertemporal hedging demand. The equity-duration channel is the clearest family-level mechanism because a rate innovation changes the discounting of cash flows at different horizons. For the other anomaly families, the paper predeclares exposure comparisons but does not force a directional sign unless the article-level result and generated portfolio definition support it.

The meaning of a numerical rate innovation can also change across regimes. Conventional positive-rate policy, lower-bound periods, asset-purchase and forward-guidance episodes, and inflation-driven tightening can attach different information to the same measured short-rate

Table 1: Replication target and economic hypotheses

Claim	Role	Primary test	Interpretation rule
R1a–R1f	Replication targets	Layer-specific article statistics within tolerances	Numerical recovery only
H1	Incremental pricing content	Improvement versus the ex ante CAPM comparator	Economic thresholds required against CAPM; the strongest observed comparator is a secondary adversarial check
H2	Temporal stability	Post-publication compatibility	Sign and magnitude matter
H3	Regime stability	Separate beta, fitted-premium, and error tests	Non-rejection is insufficient
H4a	Cross-sectional identification strength	Beta rank, standardized exposure dispersion, and the numerical spanning criterion	Weak identification blocks interpretation
H4b	Influence stability	Leave-one-family refits and standardized influence	One family or one portfolio may not drive the conclusion
H4c	Fitted-premium precision	Joint-bootstrap interval for the rate-attributable fitted premium	An interval spanning both economic directions is uninterpretable

Source: Authors' replication protocol.

movement. The baseline rate residual therefore remains an aggregate state-variable innovation; policy and information language is reserved for the optional announcement-based appendix design.

Some implementation details remain ambiguous at the source-file level. The article identifies the Federal Reserve Bank of St. Louis, Kenneth French's data library, Robert Stambaugh's web page, and Lu Zhang as sources, but the exact download files and archive vintages still have to be frozen before the manuscript can claim exact computational replication.

4 Data and replication design

The data design separates exact inputs, approximate reconstructions, and extension inputs. Exact inputs require a verified article definition, source, vintage, transformation, and estimator role. Approximate reconstructions are documented public or author-data substitutes and are never labelled as exact replication. Extension inputs support post-publication, regime, and high-frequency analyses after their source terms and transformations are frozen.

The baseline sample runs from 1972-01 through 2013-12. The temporal extension begins in 2014-01, with revised-history checks kept separate from the baseline verdict. Every source must record access status, redistribution status, and its role in strict replication before it can support a claim. Exact unavailable inputs lead to a missing-input classification rather than an empirical contradiction.

The short-rate registry fixes the documented reconstruction series before the AR model is estimated. The federal-funds reconstruction uses FRED series FEDFUNDS, an effective federal funds rate, monthly provider average, annualized percent units, no seasonal adjustment, no imputation, and a frozen retrieval vintage. The Treasury-bill reconstruction uses TB3MS as the monthly secondary-market candidate and treats DTB3 only as a sensitivity candidate, because neither the article nor the supplement names a daily source or a daily-to-monthly aggregation rule. The monthly aggregation of both series is verified rather than assumed: each provider monthly value equals the corresponding daily mean rounded to the published precision in every complete month, and two competing aggregation rules are rejected. None of these rate choices is eligible for an exact replication label, because the article names a provider without a series code or an archive vintage. Source identity, not numerical proximity, decides the replication mode.

Portfolio continuity is the largest post-publication constraint. All seven registered anomaly families are available from the author source the article names, but only in a post-publication vintage that runs to 2025-12; no publication-era vintage of those files is publicly recoverable. The extension therefore proceeds as a documented reconstruction on the author's own updated panels, and the portfolio sample rather than the rate and factor data sets the binding extension endpoint. Security-level reconstruction from CRSP and Compustat remains the only route to an exact continuation; database access is not assumed and redistribution of licensed extracts is prohibited, so that route is not taken. Portfolio catalogues from other providers are never substituted for a named author source, and no acquired portfolio supports an exact-replication claim unless source compatibility is independently verified.

Short-rate series are stored in their source units and transformed only in named innovation steps. Market and portfolio returns are aligned to the canonical monthly panel before excess returns, factor construction, or model comparison. Missing returns and shock observations are not forward filled. Comparator models use identical asset-date intersections within each comparison, and delisting or survivorship treatment is recorded from verified source documentation rather than inferred from the data file alone.

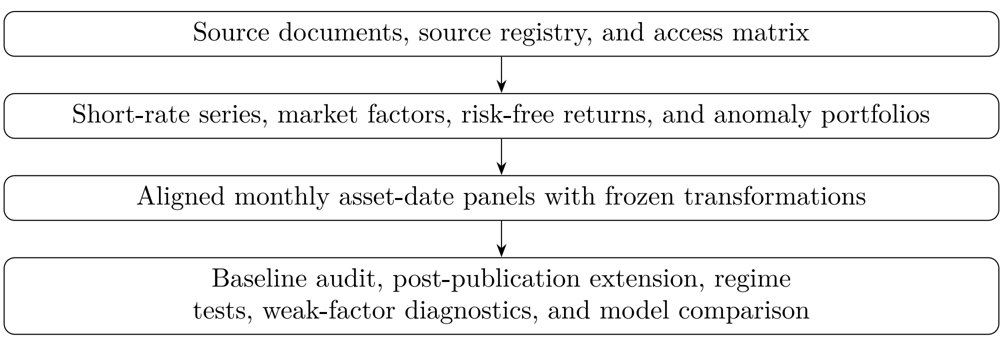

Figure 1: Data flow from sources to manuscript evidence
Source: Authors' replication protocol.

The replication target is table-level. Each published table or appendix table is mapped to a source location, model, estimator, tolerance rule, and status. A generated statistic can be labelled reproduced only after the same definition and estimator are linked to an auditable output. Nearby public data may be useful for documented reconstruction, but they cannot change the strict replication label. The replication target is decomposed into short-rate innovations, first-pass betas, cross-sectional risk prices, pricing errors and fit, comparator-model results, and supplementary robustness targets.

Table 2: Data sources and sample-construction status

Variable or family	Source	Frequency	Transformation	Status
Article and supplement	Publisher files	Document	Target extraction	Exact local access
Federal funds rate	Federal Reserve Bank of St. Louis	Monthly	FEDFUNDS documented reconstruction; first-order innovation	Acquired and frozen; aggregation verified; not exact
Treasury-bill rate	Federal Reserve Bank of St. Louis	Monthly	TB3MS documented reconstruction; DTB3 sensitivity only	Acquired and frozen; aggregation verified; not exact
Market and risk-free returns	Kenneth French Data Library	Monthly	Excess-return factors	Acquired at a publication-era and a current vintage; not exact
Anomaly decile portfolios	Lu Zhang and cited sources	Monthly	Excess returns after alignment	Acquired from the named source at a post-publication vintage; equal-weighted variants unavailable
Comparator factors	French, Stambaugh, and q-factor sources	Monthly	Common-intersection factors	All registered comparator inputs acquired and frozen; the liquidity column and scale are identified empirically and its published extremes are reproduced by no recoverable vintage
Announcement components	Documented research dataset	Event	Monthly aggregation	Extension input pending

No input is exact, because the article names providers and people rather than files. Source: `research/data_access_matrix.csv`.

5 Econometric methods

The baseline empirical design starts with the article's first-order short-rate innovation equations. Alternative autoregressive orders, alternative rate definitions, local-level filters, and announcement-surprise measures are robustness or extension specifications, not replacements for the strict baseline.

First-pass time-series regressions estimate portfolio exposures to the market excess return and the selected short-rate innovation. Their standard errors are heteroskedasticity and autocorrelation consistent, computed with the Newey and West (1987) covariance estimator under an automatic monthly lag rule that depends only on the number of observations, so no lag length is chosen after seeing an estimate. The notation separates the time-series intercept δ_i, factor loadings β_i, residuals $\varepsilon_{i,t}$, cross-sectional pricing errors α_i^{CS}, and risk prices λ. The second pass is the article's verified no-intercept OLS regression of average returns on estimated betas unless a supplement target explicitly calls for an unrestricted zero-beta-rate variant, GLS estimator, or repeated monthly cross-sectional estimator. Shanken corrections apply to the reported full-sample risk-price estimator in the benchmark design; bootstrap procedures are labelled by resampling unit before any p-value is interpreted.

Model comparison reports pricing-error loss and fit on common asset-date intersections. The registered outcomes are root mean squared pricing error, mean absolute pricing error, maximum absolute pricing error, cross-sectional fit, and joint pricing-error tests when the estimator supports them. Statistical significance is not enough: every supported result must also clear the registered economic-magnitude criterion. For H1, materiality is defined first against the ex ante CAPM comparator on the common intersection. The short-rate model must reduce RMSE and MAE by at least 10 percent and reduce the maximum absolute pricing error by at least 0.25 monthly percentage points relative to CAPM. The strongest observed registered non-short-rate comparator is retained only as a secondary adversarial analysis with model-selection uncertainty.

Weak-factor diagnostics evaluate whether short-rate betas contain enough cross-sectional variation to support a pricing interpretation. They are split into three separately classified confirmatory claims rather than one composite gate, because identification strength, influence stability, and estimation precision fail for different reasons and carry different remedies. Raw beta dispersion, singular values, and condition numbers may be reported descriptively, but they are not separate confirmatory failure rules. A short-rate risk price that fails any of the three is interpreted as weakly identified even when fit metrics appear favorable.

H4a is cross-sectional identification strength. It fails when the beta matrix loses rank, when standardized rate-exposure dispersion is less than 10 percent of standardized market-exposure dispersion, or when the numerical spanning criterion fails. Spanning is decided mechanically: the short-rate innovation is regressed by OLS on a constant and the registered non-short-rate comparator factors available on the same intersection, and the gate fails when the spanning R^2 exceeds 0.90, equivalently when the residual standard deviation falls below $\sqrt{0.10}$ of the raw factor standard deviation. The R^2 form is authoritative and the residual cutoff is the exact square root rather than a decimal rounded below it, so the two forms agree on every boundary case. Both statistics are scale free. H4b is influence stability, which fails when a leave-one-family refit reverses the sign or materiality of the rate-attributable fitted-premium spread or when a single portfolio reaches a standardized influence of one on the rate risk price. H4c is fitted-premium precision, which fails when the 95 percent joint-bootstrap interval for the fitted-premium spread contains both $+0.25$ and -0.25 monthly percentage points.

Temporal and regime methods are designed as extensions. The post-publication analysis has frozen-parameter and refitted variants, with no-lookahead rules for factor construction, portfolio membership, model vintage, and sample intersections. Regime analysis separates four questions:

stability of the rate innovation construction, stability of portfolio betas, stability of cross-sectional risk prices, and stability of pricing errors. Pooled time-series interactions test beta stability; regime-specific or locally estimated second-pass models test risk-price stability. Stability is not inferred from a failure to reject equality.

Regime risk-price stability is evaluated through the rate-attributable fitted premium $\pi_{i,s} = \beta_{i,FFR,s}\lambda_{FFR,s}$, not through an absolute bound on $\lambda_{FFR,s}$. This object is invariant to rescaling the short-rate innovation. The predeclared equivalence bounds require fitted-premium changes no larger than 0.25 monthly percentage points, fitted-premium dispersion changes no larger than 25 percent, RMSE deterioration no larger than 10 percent, maximum pricing-error deterioration no larger than 0.25 monthly percentage points, and selected-fit deterioration no larger than 0.10 under the fit contract.

Cross-sectional fit is mechanical rather than discretionary. The primary fit statistic is the article's exact definition if verified; otherwise the fallback is a no-intercept pricing-error pseudo-R^2, $1 - \sum_i (\alpha_i^{CS})^2 / \sum_i (\bar{R}_i^e)^2$, reported only when the denominator is positive and all models use the same asset-date intersection.

Inference uses Shanken-adjusted covariance for the benchmark full-sample second-pass risk-price estimator. Fitted-premium uncertainty uses a joint moving-block bootstrap with 10,000 repetitions, a Politis-White block-length selector, and a fixed 12-month fallback under declared selector failures. The stochastic variables resampled jointly by calendar month are the market factor, the comparator factors, the portfolio returns, and the short-rate level series. Regime labels are not among them: a label is a deterministic calendar function of the observation month, so each resampled month carries its own frozen label as a fixed attribute. Resampling labels would fabricate uncertainty about a known policy calendar. Each draw re-estimates the rate innovation, first-pass betas, second-pass risk prices, fitted premia, pricing errors, fit metrics, and equivalence statistics. Equivalence claims use one rule: standard two one-sided tests at the 5 percent level, implemented as inclusion of the two-sided 90 percent bootstrap interval inside the declared bound. The stricter 95 percent interval-inclusion variant is reported only as a labelled sensitivity column and never as the confirmatory test.

The primary regime registry contains six regimes: conventional positive-rate policy, effective lower bound and first QE, first normalisation, pandemic lower-bound and QE, inflation-driven tightening, and post-tightening easing and normalisation. The post-2022 period is split because a rapid tightening cycle and the subsequent reduction from the cycle peak attach different information to the same measured rate movement; the split boundary is the first policy reduction on 18 September 2024. The primary transition rule assigns a regime to the first calendar month that lies entirely under the new policy stance, so the month containing the policy action stays with the outgoing regime. The whole-transition-month rule, which assigns the action month to the incoming regime, is preserved as a declared sensitivity analysis and never decides a confirmatory classification. Boundary-shift checks remain plus or minus three months. The entire pre-lower-bound sample is defensible as the confirmatory article-era regime because the federal funds rate is the primary short-rate policy instrument away from the lower bound; finer pre-lower-bound breaks remain exploratory and cannot replace the confirmatory calendar regime.

Regime-estimation eligibility is frozen numerically before any regime estimate is produced and is not relaxed for regimes that turn out to be short. A regime shorter than 36 months supports no regime-specific estimation and enters only pooled regime-interaction models. A regime of 36 to 71 months supports regime-specific first-pass betas under a short-sample flag but no standalone second pass. A standalone regime-specific second pass additionally requires at least 72 months, at least 10 test assets, and a beta matrix whose rank equals the number of priced factors. Under these frozen rules only the conventional and lower-bound regimes carry full eligibility, and the three post-

2020 regimes are restricted to pooled interactions; a predeclared combination of the two post-2022 regimes is available as a first-pass sensitivity.

Table 3: Predeclared monetary regimes

ID	Regime	Start	End	Obs.	Primary instrument	Eligibility
`conventional_pre_elb`	Conventional positive-rate policy	1972-01	2008-12	444	Federal funds rate	First pass and standalone second pass
`elb_qe`	Effective lower bound and first QE	2009-01	2015-12	84	Target range and asset purchases	First pass and standalone second pass
`normalisation`	First normalisation	2016-01	2020-03	51	Target range and balance-sheet normalisation	First pass with short-sample flag; standalone second pass blocked
`pandemic_elb_qe`	Pandemic lower-bound and QE	2020-04	2022-03	24	Lower-bound target range and asset purchases	Below the 36-month floor; pooled interactions only
`inflation_tightening`	Inflation-driven tightening	2022-04	2024-09	30	Federal funds target range	Below the 36-month floor; pooled interactions only
`post_tightening_easing`	Post-tightening easing and normalisation	2024-10	2026-06 registered, 2025-12 effective	21 registered, 15 usable	Federal funds target range	Below the 36-month floor; pooled interactions only

A regime begins in the first calendar month lying entirely under the new policy stance, so the month containing the policy action stays with the outgoing regime; the whole-transition-month rule is a declared sensitivity. Permitted boundary shifts are plus or minus three months. Registered observation counts are those frozen in the regime registry, which ends the final regime at the rate-data freeze month 2026-06. Regime estimation uses the registered anomaly decile portfolios, which end 2025-12, so the last six months of the post-tightening easing window carry no test-asset observations and the usable count for that regime is 15 rather than 21. The declared post-2022 combination is affected in the same way and has 45 usable months rather than 51. The registry rows are not restated here; the effective endpoint is the binding constraint on every regime estimate. Source: `research/regime_registry.csv`, `research/anomaly_portfolio_availability.csv`, `configs/extensions.yaml`.

The high-frequency decomposition is optional rather than part of the main contribution. If included, it starts from high-frequency components constructed under an explicit announcement-based identification design after source terms and component definitions are frozen. The monthly aggregation can create a sparse factor with many no-event months, so the component factors must pass their own weak-factor diagnostics before entering the main argument.

Out-of-sample evaluation defines the forecast origin, training window, refit schedule, benchmark models, loss functions, and interpretation before results are inspected. It is an appendix robustness exercise unless it produces evidence strong enough to motivate a separate contribution. Multiplicity adjustments are predeclared within replication, baseline pricing, temporal extension, regime stability, and weak-factor families.

6 Baseline reconstruction

6.1 The short-rate innovation

The first layer to reproduce is the state variable itself. Estimating the article's autoregression on independently acquired federal funds and Treasury-bill series over the baseline window gives the coefficients in Table 5. The persistence parameters agree with the published values at printed precision for both series, and the fit statistics agree to two decimals.

Table 4: Method-to-claim map

Claim	Primary method	Decision rule
R1a–R1f	Layer-specific comparison to registered article statistics	Recovery within declared tolerances
H1	No-intercept two-pass pricing and model comparison	Must beat ex ante CAPM by materiality thresholds; strongest observed comparator is secondary
H2	Frozen and refitted post-publication evaluation	Sign, magnitude, and loss compatibility must hold
H3	Separate beta, fitted-premium, and pricing-error stability tests	Estimates must remain inside numerical bounds under 5 percent TOST
H4a	Beta rank, standardized exposure dispersion, and spanning regression	All three identification gates must pass
H4b	Leave-one-family refits and standardized influence diagnostics	No sign reversal, no materiality loss, and no dominant portfolio
H4c	Joint moving-block bootstrap of the rate-attributable fitted premium	The interval may not span both economic directions

Source: Authors' replication protocol.

One conversion is required before the intercepts can be compared. The article prints its autoregressive intercept as 0.000 while printing the innovation descriptives of its Table 1 in percent, so the intercept is stated in decimal rate units and must be divided by one hundred before comparison. Applying that conversion, both reconstructed intercepts round to the published 0.000; without it, neither does. The slope, both t-ratios, and the coefficient of determination are scale invariant and need no conversion.

Table 5: Short-rate innovation reconstruction, 1972-01 to 2013-12, under the recovered pre-window-lag timing convention. Published values are those printed by Maio and Santa-Clara (2017). The intercept is in decimal rate units and the innovation standard deviation in annualized percentage points.

Series	Intercept	Slope	Slope t	R^2	Innovation s.d.
Federal funds, reconstructed	0.000463	0.9905	147.27	0.9774	0.5855
Federal funds, published	0.000	0.991	—	0.98	0.59
Treasury bill, reconstructed	0.000354	0.9916	153.18	0.9791	0.4855
Treasury bill, published	0.000	0.992	—	0.98	0.49

The timing convention is not stated in the article and had to be recovered. Two variants are admissible: regressing the first window month on the level of the month preceding the window, or beginning the regression one month inside the window. Only the first recovers the published slopes, t-ratios, and fit statistics, and it is the variant used throughout. Under it the federal funds reconstruction matches nine of ten registered Table 1 statistics and the Treasury-bill reconstruction matches ten of ten.

That recovered convention does not carry the pricing result, which the article's own supplement suggests checking and which matters more here than there, because the convention had to be inferred rather than read. Replacing the AR(1) residual with the first difference of the rate, which has no timing convention to recover, leaves the joint seventy-portfolio estimate at -0.6863 with a nominal Shanken t of -2.87, against -0.6951 and -2.85 for the AR(1) residual on the same months. The Treasury-bill versions agree as closely, and across the thirty-two systems formed by the two rates, the two definitions and the eight asset sets, thirty carry a negative and significant rate price. The choice this reconstruction had to make is therefore not the choice the finding rests on.

Nor is the result an artefact of the years the extension later finds the relation deteriorating in. The article's supplement also reports a sample ending in December 2006, and on those 420 months the funds-rate price is −0.7329 with a nominal Shanken t of −2.97 and a fit of 0.6097, against −0.6985, −2.86 and 0.5439 on the full baseline. Dropping the financial crisis and the first effective lower bound strengthens the relation rather than weakening it, which is what makes the post-2013 deterioration reported in Section 8 a statement about the later period rather than about the estimator.

The supplement does not say whether its restricted-sample AR(1) is re-estimated on the shorter window or carried over from the full one, an ambiguity this project recorded before running either. Both are admissible and both are run rather than one being chosen silently: they give −0.7329 and −0.7354, so the undetermined convention is immaterial here, which is a stronger statement than picking a side would have been.

Neither series earns an exact-replication label. The article names its providers without giving series codes or archive vintages, so no reconstruction of the rate is eligible for that label at any recovery rate, and both are classified `approximately_reproduced_under_documented_reconstruction`.

6.2 Cross-sectional pricing

Table 6 reports the second pass for all eight registered models on the joint seventy-portfolio system over the 504 baseline months.

Table 6: Baseline cross-sectional pricing, joint seventy-portfolio system, 1972-01 to 2013-12, 504 months. Risk prices are monthly percentage points with Shanken t-ratios beneath. Fit is the article's centred cross-sectional variance ratio. All rows carry the label `documented_reconstruction`.

Model	λ_{RM}	λ_{rate}	RMSE	MAE	max $\lvert\alpha\rvert$	Fit	χ^2 (p)
CAPM	0.648	—	0.1893	0.1466	0.4752	−0.613	99.57 (0.009)
	(3.12)						
Market + funds-rate	0.601	−0.698	0.1004	0.0816	0.2388	0.544	35.65 (1.000)
	(2.87)	(−2.86)					
Market + bill-rate	0.620	−0.482	0.1084	0.0858	0.2781	0.468	46.77 (0.977)
	(2.97)	(−3.02)					
Fama-French three	0.612	—	0.0798	0.0630	0.2409	0.712	82.31 (0.098)
	(2.95)						
Carhart four	0.607	—	0.0720	0.0560	0.2427	0.765	66.82 (0.449)
	(2.93)						
Fama-French five	0.612	—	0.0745	0.0595	0.1743	0.749	78.14 (0.127)
	(2.95)						
q-factor	0.608	—	0.0832	0.0664	0.2164	0.687	81.45 (0.095)
	(2.94)						
Liquidity	0.615	—	0.0796	0.0623	0.2354	0.713	81.77 (0.091)
	(2.97)						

The qualitative published pattern is recovered. Adding the funds-rate innovation to the market model lowers cross-sectional RMSE from 0.1893 to 0.1004 and raises the article's fit statistic from −0.613 to 0.544, and the rate carries a negative price of risk with a nominal Shanken t of −2.86.

The Treasury-bill specification gives the same sign and a comparable t-ratio, so the finding does not hinge on which short rate is used.

The comparators' in-sample advantage does not survive the article's own misspecification check, which is worth reporting because it cuts against this paper's headline comparison rather than for it. The Internet Appendix re-runs the second pass with an unrestricted intercept and reads that intercept as the zero-beta rate; a rate significantly above the average bill return indicates misspecification. The appendix reports the outcome in prose without tabulating it, naming the CAPM, the Fama-French three-factor model, the liquidity model and the q-factor model as rejected while its own model passes.

That set is a prediction, and the reconstruction meets it exactly. On the joint seventy portfolios the excess zero-beta rate is significant for those four models and for no other, and the CAPM's estimate of 1.1417 per month matches the appendix's description of values above one percent. Both versions of the short-rate model pass, at 0.3204 with a t of 0.79 for the funds rate and 0.1725 with a t of 0.42 for the bill rate. The two comparators the appendix does not name, Carhart and the five-factor model, are also insignificant here, which is consistent with its "in most cases" and is not a claim it makes.

The reading this supports is narrow. It does not rescue the short-rate model on in-sample pricing error, where the comparators remain ahead, and a zero-beta rate indistinguishable from the bill return is a weak form of correct specification. What it does show is that the comparison the incremental-pricing standard turns on is not one-directional: the specifications that price the cross-section more accurately are the same ones this test rejects.

The article's own supplement offers a second fit measure, and computing it here shows why the paper reports the first. Its Internet Appendix replaces the variance in the denominator of equation (6) with the cross-sectional second moment, which bounds the statistic below at zero where equation (6) can be negative. The appendix warns in the same paragraph that the replacement buys that bound at a price: a model can score well on it "just by fitting well the cross-sectional mean despite not explaining any cross-sectional dispersion in risk premia". On these systems the warning is not hypothetical. Eight of the sixty-four price the cross-section worse than a constant does, which is what a negative equation (6) means, and every one of the eight scores positive on the alternative measure. The CAPM on the joint seventy portfolios moves from -0.6125 to $+0.9170$. Reporting the second measure alone would describe a model that fails as one that explains most of the cross-section, so both are generated and the paper's own metric is the one its claims rest on.

Two features of the table qualify that reading, and both work against the short-rate model. First, every registered traded multi-factor comparator attains a lower in-sample cross-sectional RMSE and a higher fit than the short-rate specification: the Carhart model reaches an RMSE of 0.0720 against 0.1004.

That comparison is stated as an in-sample one deliberately, and it does not support a ranking of the models. The comparators carry three to five factors against the short-rate model's two, and a larger factor set has more freedom to span a fixed cross-section of seventy average returns; because the first-pass betas are re-estimated for each model, this is not a nested comparison in which the usual degrees-of-freedom correction applies cleanly either. Raw RMSE is therefore not a complexity-neutral criterion. What the table establishes is that the short-rate specification, while a large improvement on the market model alone, does not achieve a smaller in-sample pricing error than any registered traded alternative on these test assets. Establishing that it is worse as a pricing model would require a genuinely predictive comparison or a complexity-adjusted criterion, neither of which this paper runs; the out-of-sample design in Appendix E is where that question belongs.

Second, the specification test is not evidence in the short-rate model's favour despite its p-value of essentially one. The Shanken multiplier for that model is 2.42, against 1.02 for the market model and between 1.06 and 1.11 for the traded-factor models. The multiplier scales the estimated covariance of the pricing errors, so a model with a large price of risk relative to factor volatility inflates the variance against which its own pricing errors are judged, and the statistic falls. The non-rejection reflects that inflation as much as it reflects small pricing errors, which is why the comparison above rests on RMSE rather than on χ^2.

6.3 What the cell-level audit recovers

Comparing the reconstruction against the published cells gives a recovery rate that varies sharply and informatively by statistic. Of 215 unique published cells, all 215 were compared, and 53 fall inside the published rounding. Market risk prices recover well, with 20 of 29 cells inside the published rounding and all 29 agreeing in sign. Rate risk prices recover in sign and magnitude but less often at printed precision, with all 16 cells negative as published and a median absolute difference roughly four times that of the market factor. The χ^2 statistic recovers least well, with none of 29 cells inside the published rounding.

That ordering is what the estimator predicts. The specification statistic inverts a seventy-by-seventy pricing-error covariance through a pseudo-inverse, which amplifies small differences in the residual covariance, so a portfolio difference that moves a risk price by 0.003 can move the statistic by a whole unit.

The article's Table 5 decomposition is audited on the same terms. It reports, for the extreme decile of each anomaly, the average excess return, the premium attributable to each factor as the loading times that factor's estimated price of risk, and the pricing error left over. Reconstructing it needs no new estimation: the loadings, the risk prices and the pricing errors are already stored, and the decomposition is arithmetic on them. Of its 84 published cells, 21 fall inside the published rounding and all 84 agree with the article in sign. That is the highest sign agreement of any table audited here, and it is the level at which the article's own argument is conducted: the economic reading it draws from that table, that the rate factor rather than the market factor accounts for the decile spreads, reconstructs on every one of the seven families.

The common source is the portfolio vintage. The anomaly decile panels come from the original author's source but in a later rebuild, and none of the seven families matches its published descriptive row on all five statistics. A cross-sectional estimator applied to slightly different portfolios returns slightly different cells. Nothing in the audit separates that explanation from an estimator discrepancy, and we do not claim otherwise. Accordingly the layers are classified as partially recovered under documented reconstruction rather than as reproduced or contradicted; a contradiction label would require a completed attempt on source-compatible inputs, which these are not.

6.4 The article's empirical p-values

Every p-value the article prints comes from the 5,000-replication useless-factor bootstrap of its Internet Appendix Section 4, in which the factors are resampled on a time sequence drawn independently of the return residuals so that they cannot explain returns by construction. Earlier drafts of this audit left those 118 cells uncompared, because the only alternative was to hold a simulated p-value against an asymptotic one, which is a different object. The procedure is now implemented and run for all 29 systems the article prints a p-value for, at the published replication count, with no degenerate draws.

One adjustment separates this comparison from the others. A bootstrap p-value is a Monte Carlo quantity whose own standard error is $\sqrt{p(1-p)/B}$, about 0.007 near $p = 0.5$ at $B = 5{,}000$, while a p-value printed to three decimals carries a rounding tolerance of 0.0005. Across most of the range the procedure's sampling noise is an order of magnitude wider than the band a cell would have to land in, so a miss carries no information about the reconstruction: the article's own bootstrap, rerun on the article's own data under a different seed, would miss its published value just as often. We therefore report those cells as not resolvable at the published precision rather than as not recovered, the same distinction drawn in Section 9 between an inconclusive result and a demonstrated difference.

On that reading, 36 of the 118 cells land inside the published rounding, 8 miss it where the tolerance is attainable, and 74 are not resolvable. Of the 37 cells whose tolerance is attainable at all, 29 are recovered.

The digits agree less than the verdicts do, and the verdicts are what the p-values were printed to support. All 45 risk-price cells agree with the article on significance at the 5 percent level, and 27 of 29 agree for each of the specification test and the cross-sectional fit. This is the strongest correspondence anywhere in the audit, and it is a correspondence of inference rather than of arithmetic.

Two limits belong with that result. The bootstrap tests a useless-factor null rather than a zero risk price under a correctly specified model, so a small p-value says the factor is not behaving like noise, not that the model is right. And the procedure is an audit instrument: it enters no registered gate, and the confirmatory inference throughout this paper remains the moving-block bootstrap of Section 5.

The baseline analysis uses the strict monthly panel, the registered baseline and comparator system, the article's two-pass estimator with the Shanken correction, and the seventy registered anomaly portfolios. Uncertainty is the Shanken covariance applied to both risk prices and pricing errors; the economic magnitude is a reduction in cross-sectional RMSE of 47.0 percent relative to the market model.

7 Weak-factor and model-comparison diagnostics

Two questions govern whether the baseline result may be interpreted at all. The mechanical one is whether estimated short-rate betas vary enough across assets for the second pass to identify a price of risk. The economic one is whether the factor reduces pricing errors by a material amount against comparator models chosen in advance. The first is answered affirmatively; the second is not.

7.1 The factor is identified in the cross-section, but its betas are noisy

Two distinct properties are at stake here and the paper keeps them apart, because its own evidence separates them. The first is whether the estimated exposures vary enough across assets, and are distinct enough from the traded factors, for the second pass to identify a price of risk at all. They do. The second is whether an individual estimated beta is a reliable measurement of that asset's exposure. Largely they are not, and Section 9.6 quantifies how far from reliable.

All three registered identification gates pass. The beta matrix attains full rank. The standardized rate-exposure dispersion is 0.2540 of the standardized market dispersion against a floor of 0.10, and per family it ranges from 0.1903 to 0.3894, so no family sits near the floor. The spanning regression of the rate innovation on all ten registered non-short-rate comparator factors returns a residual standard-deviation ratio of 0.9716 against a required minimum of $\sqrt{0.10} \approx 0.3162$: the rate innovation is a long way from being spanned by the existing traded-factor set.

Influence is likewise stable. Across the seven leave-one-family refits the rate price of risk moves within $[-0.7519, -0.6777]$ with no sign reversal, and the largest standardized influence of any single portfolio on it is 0.0896 against a bound of one.

The registered precision gate passes for every family, and its content is narrower than the word precision suggests. The gate asks whether the bootstrap interval for a family's rate-attributable fitted-premium spread accommodates economically large effects of *both* signs, that is whether it contains both $+0.25$ and -0.25. None does, so all seven pass.

That is an economic-exclusion criterion, not a statement about statistical significance, and it must not be read as one. Five of the seven intervals contain zero, so for those five the sign of the spread is not established. Only book-to-market, at $[0.0223, 0.8321]$, and investment-to-assets, at $[-0.5342, -0.0239]$, exclude zero. We report the gate as registered and decline to restate it as sign identification, which it never was.

Intervals come from a moving-block bootstrap with 10,000 repetitions in which each draw resamples calendar months jointly and recomputes the entire chain: the autoregression, all seventy first-pass betas, the no-intercept second pass, and the fitted premia. The block length of six months was selected by the Politis-White rule rather than set by hand.

7.2 What the identification gates do and do not license

Together the gates say the short-rate factor is distinct from the traded factor set, that its estimated exposures are dispersed enough across assets for a second pass to run, that the risk price is not the artifact of any one anomaly family, and that no family's fitted-premium spread is consistent with large effects in both directions at once. That is enough to take the cross-sectional estimate seriously as an estimate.

It is not enough to treat an individual estimated beta as a measurement. The simulation in Section 9.6, calibrated to this process, attributes 61.4 percent of the cross-sectional dispersion of estimated rate betas to first-pass estimation error even at 648 months, against 2.7 percent for market betas, a reliability ratio of 0.386 against 0.973. The dispersion gate is satisfied by dispersion that is substantially noise, and the gate was never designed to distinguish the two. We flag this as a limitation of the registered identification standard rather than a finding against the factor: a scale-aware dispersion floor is a test of whether a second pass can be run, not of how well its inputs are measured.

The consequence runs through the rest of the paper. It is why the regime contrasts in Section 9 are underpowered at feasible sample lengths, and it is the most plausible reconciliation of a risk price that is significant in the full sample with economic quantities that are unstable across subsamples.

7.3 The incremental-pricing standard is not met

The materiality standard was registered before any estimate existed, with the CAPM named in advance as the primary comparator and all three gates required to hold. Table 7 reports them.

Table 7: Incremental-pricing materiality against the ex ante CAPM comparator, joint seventy-portfolio system. All three gates must hold for the claim to be supported; the registered classification is `unsupported`.

Gate	Threshold	CAPM	Market + rate	Result
RMSE reduction	$\geq 10\%$	0.1893	0.1004	pass, 47.0%
MAE reduction	$\geq 10\%$	0.1466	0.0816	pass, 44.3%
max $\lvert\alpha\rvert$ reduction	≥ 0.25 pp	0.4752	0.2388	**fail**, 0.2364 pp

The registered classification is therefore *unsupported*, and the classification being binary is no reason for the economic reading to be. The two relative gates clear by roughly a factor of four. The absolute gate fails by 0.0136 monthly percentage points. The substantive finding is not that the short-rate factor adds no incremental pricing content, which the table plainly contradicts, but that its improvement over the market model falls just short of the definition of economically sufficient that we fixed in advance, on one unusually demanding gate. We report the classification as registered and the magnitude alongside it, and neither should be quoted without the other.

That asymmetry deserves to be on the record, because it alone determines the headline classification. The market model's own maximum absolute pricing error on this asset set is 0.4752 monthly percentage points, so demanding an absolute reduction of 0.25 demands cutting that statistic by 52.6 percent. The short-rate model cuts it by 49.7 percent. In relative terms the third gate is a five times stricter bar than the first two, which ask for 10 percent. Nothing in the registered thresholds states that the asymmetry was intended. It was nonetheless fixed before any estimate existed and is applied exactly as fixed; adjusting it now, in the knowledge that it is the binding constraint, would not be defensible. We record it as a design limitation in Section 10 rather than revising it.

Two of the seven families, equity duration and investment to assets, are supported under both the funds-rate and the bill-rate specifications. The other five and the joint system are not, and the absolute gate is the binding one in six of the eight asset sets.

A secondary comparison against the strongest observed non-short-rate comparator, selected *after* observing baseline RMSE and therefore carrying model-selection uncertainty, is unsupported in all sixteen rows with every gate failing. This selection never feeds the choice of primary comparator, and it is reported because omitting it would overstate the short-rate model's standing.

These diagnostics use the strict baseline panel, the registered robustness family, the article's two-pass estimator with bootstrap and influence diagnostics, and the seventy registered anomaly portfolios. Uncertainty is governed by the weak-factor gates and the joint moving-block bootstrap; the economic magnitude is a shortfall of 0.0136 monthly percentage points on the single binding materiality gate.

8 Post-publication evidence

The baseline endpoint remains 2013-12 and the extension begins in 2014-01, so no revised datum and no later observation can change the replication label of Section 6.

8.1 Separating revision from the passage of time

An extension must use the current factor vintage while the baseline uses the publication-era vintage, so a naive comparison would confound a change over time with a change in data vintage. We therefore estimate a third system on the *baseline months* using the *current* vintage. Differencing it against the locked baseline isolates the vintage contribution; differencing the refitted extension against it isolates the temporal contribution. The registered gates are evaluated on the second comparison.

That does not mean revised data are absent from the temporal verdict, and it would be wrong to say so: the revised-history system *is* the comparator against which two of the three gates are evaluated, so revised values enter the reference quantity directly. What the construction achieves is narrower and still worth having. Both sides of the temporal comparison are built on the same vintage, so the revision contribution is common to them and differences out, leaving the change between the historical and extension windows separable from the publication-era-to-current revision that the locked-baseline comparison reports on its own.

A fourth system applies the 2013-12 parameters unchanged to the later months without re-estimating anything, and is genuinely out of sample.

Table 8: Temporal evaluations. The locked baseline and revised history share months and differ in vintage; the revised history and refitted extension share vintage and differ in months, so differencing isolates one change at a time.

Evaluation	Months	λ_{RM}	λ_{rate}	RMSE	Fit
Locked baseline, 1972–2013	504	0.6012	−0.6985	0.1004	0.544
Revised history, same months	504	0.6030	−0.6985	0.1004	0.544
Frozen parameters on 2014–2025	144	0.6012	−0.6985	0.4463	−0.691
Refitted extension, 2014–2025	144	0.9972	−0.0825	0.1918	0.450

The separation is clean, and cleaner than we first measured it. Recomputing the baseline window on a vintage nine years later moves the rate price of risk by −0.0000 and cross-sectional RMSE by −0.003 percent. Moving to the post-publication window moves them by 0.6159 and 91.0 percent. The vintage contribution is negligible on every dimension, so the post-2013 change cannot be attributed to revised historical data.

The vintage contribution is this small for a reason worth stating, because a reader who expected revisions to matter should know why they do not here. The federal funds series is not revised, and both panels read the same frozen file, so the rate innovation is identical across the two vintages by construction. The test assets are the same portfolio vintage in every window. What actually differs between the locked baseline and the revised history is the market and risk-free factors, and over these months that difference does not move the cross-section measurably. An earlier version of this comparison reported a vintage contribution of 0.0011 on the rate price; that figure was an artifact of a timing error in the revised-history autoregression, not a vintage difference, and is corrected here.

8.2 All three registered gates fail

Sign compatibility of the rate-attributable fitted-premium spreads holds for two of seven families; the magnitude gate holds for one of seven; and RMSE deteriorates by 91.0 percent against a bound of 10 percent. The registered classification is *unsupported*. Five of the seven families reverse the sign of their spread, and the two that keep it do so with spreads that have collapsed toward zero. Applying the 2013-12 model unchanged to the later months gives a fit statistic of −0.691, meaning the fitted premia explain less of the cross-section of average returns than a constant would.

Those sign reversals are weaker evidence than their count suggests, and the paper's own diagnostics say why. Section 7.1 reports that five of the seven baseline fitted-premium intervals contain zero. Of the five families whose point estimate reverses, only book-to-market had a baseline interval excluding zero; the other four reverse a sign that was never established. Counting reversals therefore overstates the evidence, and we report the count only alongside this qualification.

8.3 What the registered gates do not establish

The registered H2 rule compares point estimates against fixed bounds, so it can report compatibility or its absence but never that the data cannot resolve the question. The hypothesis registry anticipates an inconclusive state; the point-estimate rule alone cannot reach it. We therefore add a supplementary interval estimate, on the same joint moving-block bootstrap the regime analysis uses, re-estimating both windows in every draw and pairing the two independent distributions. It is reported beside the registered verdict and never replaces it.

The result changes what the temporal evidence can be said to show. Of the seven per-family fitted-premium changes, not one is resolved: every 90 percent interval straddles the 0.25 bound, so none demonstrates a change and none certifies equivalence. The point estimates are large, up to -0.9881 for earnings to price, but so are the intervals, and the largest of them spans from -1.68 to -0.06. The single demonstrated exceedance is the relative deterioration in cross-sectional RMSE, whose entire interval, $[+0.173, +1.844]$, lies above the 10 percent bound.

That pattern is the same one the regime analysis returns in Section 9, and for the same reason. What is demonstrated is a deterioration in pricing-error performance. What is not demonstrated, at this sample size, is a change of any particular size in what the factor is paid. The registered gates fail; the economic change behind them remains unresolved.

8.4 The registered dispersion gate does not account for the result

The extension window contains the effective lower bound, so the short-rate factor has far less variation in it: the innovation standard deviation falls from 0.5855 to 0.1707, and the share of months with the rate below half a percent rises from 12.3 to 41.7 percent. That invites the reading that the extension fails because the factor is no longer identified rather than because the relation changed.

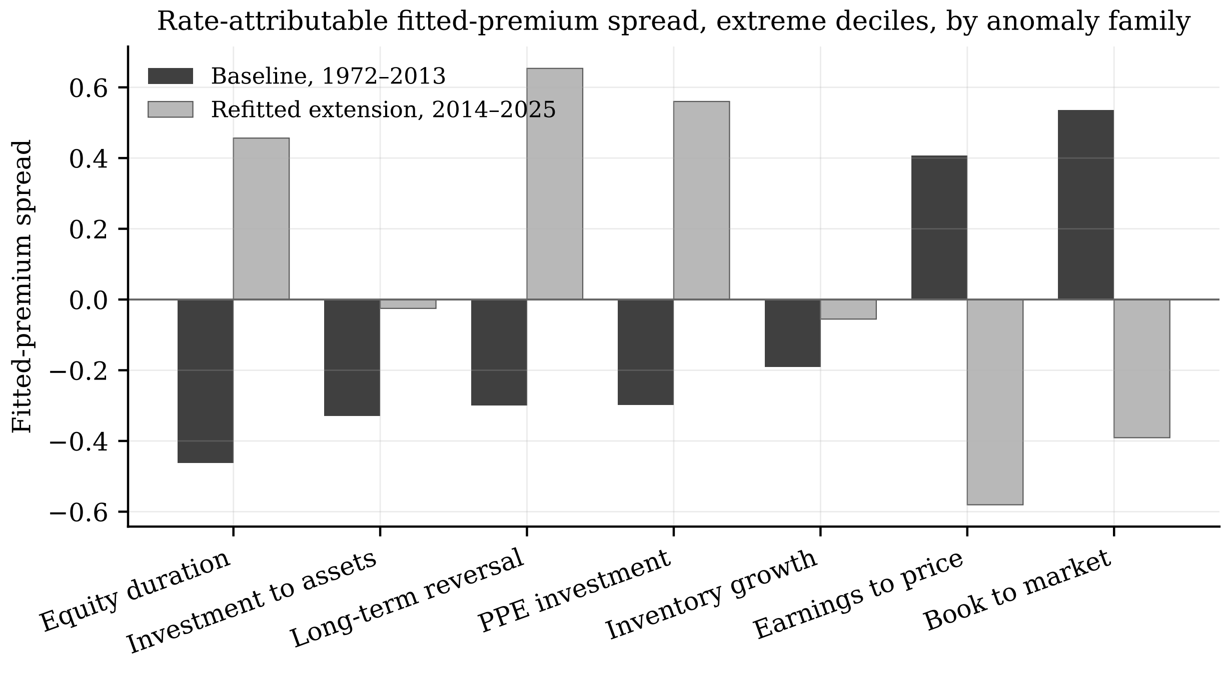

Figure 2: Rate-attributable fitted-premium spread between extreme deciles, by anomaly family, in the baseline window and in the refitted extension. Five of the seven families reverse sign; the two that keep it do so with spreads that have collapsed toward zero.

Source: `artifacts/tables/extension/fitted_premium_spreads.csv`.

The registered identification gate is not the factor's own variance but the cross-sectional dispersion of standardized exposures, which is scale aware by construction. That statistic *rises* from

0.2540 in the baseline to 0.5800 in the extension, against a floor of 0.10. Smaller factor variation is offset by larger betas, and the extension window clears the identification floor more comfortably than the baseline does.

Clearing that floor rules out the registered explanation, not the general one. The gate is a dispersion statistic, and Section 9.6 shows it is not a reliability statistic: the same rate betas carry a reliability ratio of 0.386 over the full 648 months, so most of the dispersion the gate measures is first-pass estimation error rather than spread in true exposures. A window can therefore pass H4a while its betas remain too poorly resolved to price. Passing the gate closes off the reading that the extension fails because the factor has no cross-sectional variation left; it does not close off measurement error as a contributor to the instability.

The extension estimate is nonetheless imprecise in its own right, at -0.0825 with a nominal Shanken t of -1.49 against -0.6985 and -2.86 in the baseline. A collapsed point estimate with a wide interval is consistent both with a relation that has weakened and with one estimated on 144 months instead of 504, and we do not choose between those.

The temporal analysis uses the extension-compatible monthly panel, frozen and refitted versions of the registered two-factor system, the same source-compatible anomaly portfolios in every window, and the predeclared temporal comparisons. Uncertainty is the Shanken covariance with the registered identification diagnostics; the economic magnitude is a 91.0 percent deterioration in cross-sectional RMSE against a 10 percent bound.

9 Monetary-regime stability

Pooled time-series interactions are evidence about exposure stability, not by themselves evidence about cross-sectional risk-price stability, which requires regime-specific second-pass models. The registered estimator therefore has two halves. They point the same way, but only one of them has the power to demonstrate it.

Two registered regimes straddle the 2013-12 vintage boundary, so a panel that switched vintage at its natural break would confound a policy difference with a data revision inside a single regime. The regime panel is built on the current vintage throughout, spanning 648 months from 1972-01 to 2025-12, which Section 8 licenses: over the baseline window the vintage contribution is under 0.01 percent of RMSE against temporal contributions of 91.0 percent.

9.1 Most regimes cannot support a standalone second pass

The eligibility floors were fixed before any regime was estimated. Table 9 applies them. Two of the six registered regimes clear the standalone second-pass tier. A third, the normalisation regime, sits in the short-sample band and keeps its first-pass betas while the standalone second pass stays blocked. The remaining three fall below the first-pass floor entirely.

The confirmatory equivalence comparison therefore spans a single contrast. The four ineligible regimes are not dropped: they enter through the pooled interaction model, which borrows strength across the whole sample.

9.2 The two estimable regimes

The conventional-regime estimate reproduces the baseline closely, at -0.7259 on 444 months against -0.6985 on the 504-month baseline, with the same Shanken t-ratio to two decimals. The lower-bound estimate is a different object: the price of risk is not merely smaller but indistinguishable from zero, and the fit statistic is negative. The average absolute rate-attributable fitted premium

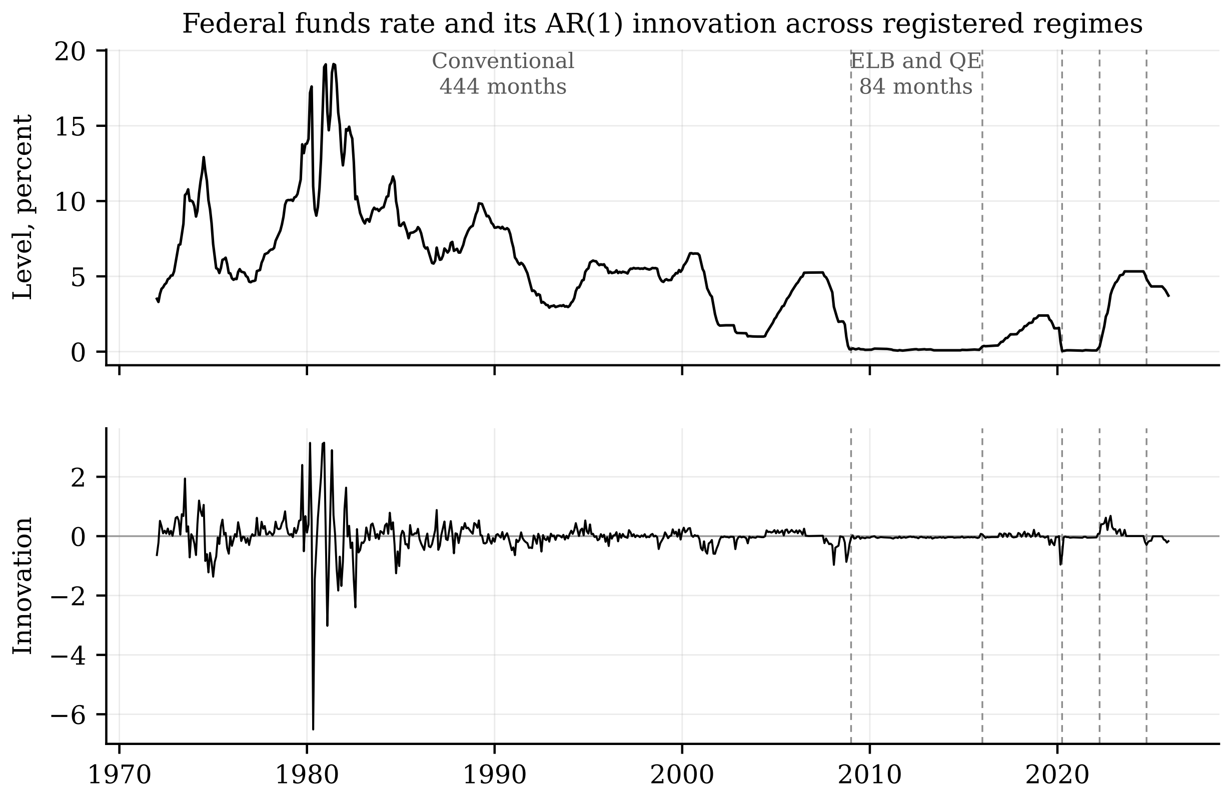

Figure 3: The federal funds rate and its autoregressive innovation across the registered regimes, on the vintage-consistent panel. Dashed lines mark regime boundaries. The innovation that identifies the factor retains almost none of its earlier variation once the policy rate reaches the effective lower bound.

Sources: `artifacts/tables/regimes/regime_monthly_series.csv` and `artifacts/tables/regimes/regime_eligibility.csv`.

Table 9: Registered monetary regimes and their estimation eligibility under the frozen floors. Regimes without a standalone second pass enter the analysis only through the pooled interaction model.

Regime	Window	Months	Second pass	First pass
`conventional_pre_elb`	1972-01 to 2008-12	444	yes	yes
`elb_qe`	2009-01 to 2015-12	84	yes	yes
`normalisation`	2016-01 to 2020-03	51	no	yes
`pandemic_elb_qe`	2020-04 to 2022-03	24	no	no
`inflation_tightening`	2022-04 to 2024-09	30	no	no
`post_tightening_easing`	2024-10 to 2025-12	15	no	no

Table 10: Regime-specific second passes on the joint seventy-portfolio system, current vintage throughout. Only regimes clearing the frozen eligibility floors appear. Dispersion is the cross-sectional standard deviation of rate-attributable fitted premia.

Regime	Months	λ_{RM}	λ_{rate}	Shanken t	RMSE	Fit	Dispersion
Conventional, 1972–2008	444	0.458	−0.7259	−2.87	0.1046	0.596	0.1753
Lower bound, 2009–2015	84	1.327	0.0037	0.84	0.2220	−0.172	0.0333

falls from 0.1517 to 0.0281 monthly percentage points, and every one of the seven families moves toward zero rather than reversing.

One regime can be estimated outside the confirmatory family, and it is worth reporting because it is the only look available at the years between the lower bound and the pandemic. The normalisation regime has 51 months against 70 test assets, so its residual covariance is rank deficient, at rank 48. That makes it ineligible under the frozen floor but not incomputable: no step of the second pass inverts the residual covariance, and the pricing-error test already uses a pseudo-inverse. Estimated on that basis it gives a rate price of -0.0844 with a nominal Shanken t of -2.14, between the conventional regime's -0.7259 and the lower bound's 0.0037.

We report it as a sensitivity and nothing more. It enters no registered classification, no multiplicity family, and not the H3 verdict; its short sample carries the registered flag; and its chi-square is unreadable, since the statistic is referred to $N - K$ degrees of freedom while its pseudo-inverse measures at most 48 directions. The frozen floor is unchanged. What the estimate adds is a datum consistent with the rest of the section rather than a new claim: the rate price after 2008 is small in every window we can look at, and small differently in each.

9.3 Equivalence is not demonstrated, and neither is difference

The confirmatory rule is two one-sided tests at the 5 percent level, implemented as inclusion of the two-sided 90 percent joint-bootstrap interval inside the declared bound, applied to the rate-attributable fitted premium of every one of the seventy test assets. All five registered gates fail to certify equivalence.

An equivalence test can fail two ways, and only one of them is a finding. The interval can lie wholly beyond the bound, which is evidence of a real difference, or it can straddle the bound, which means the data cannot resolve the question. Sorting the seventy per-portfolio tests: 26 certify equivalence, 44 are inconclusive, and *none* demonstrates a change exceeding the registered bound of 0.25 monthly percentage points. Sixteen have point estimates outside the bound, the largest being 0.5272, but in every case the interval reaches back inside it.

Among the aggregate gates the pattern is the same but for one exception. The relative deterioration in cross-sectional RMSE has its entire 90 percent interval above the bound, at $[+11.4, +135.9]$ percent against 10 percent; the dispersion, maximum pricing error and fit comparisons are all inconclusive.

The registered outcome is *unsupported*, and the wording matters. That label asserts regime dependence, which is a positive claim, so we assign it only because one dimension demonstrates an exceedance. Had that gate also been inconclusive the registered outcome would have been the inconclusive one, since no other equivalence dimension supplies affirmative evidence.

9.4 Exposures are regime dependent

Using all 648 months and interacting both factors with the regime label, the restriction that rate betas are equal across regimes is rejected with a statistic of 37.31 on 5 degrees of freedom, an unadjusted p-value of 5.2×10^{-7} and a Holm-adjusted p-value of 5.1×10^{-5}; the all-factor restriction is rejected more strongly still. Per asset, rate-beta interactions are significant for 26 of 70 after adjustment. Holm was applied across the 142 pooled tests, and the registered family also contains the equivalence tests above; a Bonferroni bound over the completed family of 216 leaves both joint results significant, so the conclusion does not depend on where the family boundary is drawn.

One fragility is recorded. Shifting every regime boundary by three months either way leaves the verdict intact but not the aggregate statistic that produces it: its adjusted p-value rises to 0.065 and 0.051 respectively, so the registered boundaries are the most favourable of the three for that statistic. The verdict survives on the per-asset rejections, which increase to 39 and 36.

A battery of structural-break tests is reported as exploratory and is excluded from the confirmatory family. Its most useful result is negative: left to find breaks on its own, the sample selects 1989-12, 1998-07 and 2001-08, none of them a monetary-policy date, and formal tests at the registered boundaries do not reject at the 5 percent level. The dates on which this cross-section breaks are not the dates on which monetary policy changed.

9.5 What the regime evidence establishes, in three parts

The pooled and regime-specific halves answer different questions, and reading them as one verdict would overstate both. Separating them:

1. **Rate exposures differ across the registered regimes.** This is established, by the pooled interaction test on all 648 months, and it survives adjustment over the completed multiplicity family and a three-month shift of every boundary. It is a statement about first-pass betas.

2. **Regime-specific differences in the price of risk and in rate-attributable fitted premia are not identified at feasible regime lengths.** Of the seventy per-portfolio equivalence tests, 26 certify equivalence, 44 are inconclusive, and none demonstrates a difference beyond the registered bound. This is neither evidence of stability nor evidence of change.

3. **One aggregate pricing measure deteriorates beyond its bound.** Relative cross-sectional RMSE has its entire 90 percent interval above the 10 percent bound. This is the single affirmative exceedance in the milestone and it is what carries the registered classification.

Instability of exposures does not by itself imply instability of the price of risk, and we do not claim it does. A factor whose betas move across regimes can still be priced identically in each; establishing otherwise requires the regime-specific second passes, which is exactly where the sample runs out. The break tests reinforce the caution rather than the narrative: the dates this cross-section actually breaks on are not policy dates.

9.6 Why the equivalence tests could not resolve the question

A simulation with 2,000 replications at each of sixteen window lengths asks how much sample the registered equivalence problem needs. It is a calibrated experiment, not a general property of the two-pass estimator, and its design choices all favour the estimator: factors and disturbances are independent and Gaussian, and the full-sample estimated betas are treated as true, so the process is not a fixed point of the estimator it is used to study and the simulated cross-section is better identified than the real one. Read as such, the finding is that *under this calibrated design*, samples of the observed regime lengths do not supply the precision the registered equivalence problem requires.

On the joint seventy-asset system, resolving fitted-premium spreads against the 0.25 bound requires 180 months; the lower-bound regime has 84. Driving the probability of a sign error on the rate price of risk below 5 percent requires 444 months. The design flatters the estimator in several respects, so these are best read as optimistic benchmarks rather than as precise requirements for the empirical problem. We do not claim them as lower bounds: the calibration is stylized, the true-beta distribution is itself built from estimated betas, and the simulation is not a fixed point of the estimator, so the ordering between simulated and empirical finite-sample difficulty is argued rather than established.

The mechanism is errors-in-variables attenuation. It does vanish asymptotically under the stationary model simulated here, since the estimation variance carries an explicit $1/T$, but it remains economically large at every sample length considered: even at 648 months, 61.4 percent of the cross-sectional dispersion of estimated rate betas is first-pass estimation error, against 2.7 percent for market betas. The sampling standard deviation of the rate price of risk is essentially flat, 0.1499 at 12 months and 0.1167 at 648, so a longer window moves the estimate rather than tightening it.

One by-product of the exercise deserves more weight than its role as a power calculation, because it bears on every Shanken interval in this paper. Under the calibrated design the nominal 95 percent Shanken interval for the rate price of risk never attains coverage inside $[0.90, 0.99]$ at any simulated window length. It runs from 0.227 at 12 months to 0.650 at 648, with a minimum of 0.176 at 30 months. The same interval for the market price of risk is well behaved throughout, with coverage between 0.943 and 0.959 at every length, which is the control showing the machinery itself is sound. The gap between the two is the errors-in-variables problem again: the Shanken correction adjusts for beta estimation error under assumptions that a factor with this noise share strains.

How far this carries to the empirical estimate needs care, and we deliberately stop short of the strongest reading. The calibration treats full-sample estimated betas as true and then re-estimates them with fresh noise, so the simulated system is not a fixed point of its own estimator: part of the attenuation, and part of the undercoverage, is built into that construction rather than measured from the data. We therefore describe the baseline t-ratio of -2.86 as *nominal* throughout, and treat the simulation as a warning that its nominal size may overstate the evidence rather than as a measurement of how far it does. Establishing the latter would require a latent-beta calibration that reproduces the observed cross-sectional beta dispersion after re-estimation, which we do not attempt here.

Within those limits the exercise still does useful work. It offers the most coherent account available of how a risk price that is nominally significant in the full sample coexists with economic quantities that will not hold still across subsamples, and it qualifies the identification discussion in Section 7.1: passing a dispersion floor is not the same as measuring exposures well.

The regime analysis uses the vintage-consistent 648-month panel, the predeclared regime indicators, pooled interaction tests for exposures and separate second-pass models for prices of risk, and the same seventy anomaly portfolios throughout. Uncertainty comes from Holm-adjusted Wald tests and the joint moving-block bootstrap; the economic magnitude is a fall in the average absolute rate-attributable fitted premium from 0.1517 to 0.0281 monthly percentage points between the two estimable regimes.

10 Discussion and limitations

The first limitation is the input layer, and it bounds every claim in the paper. The article names its providers without series codes or archive vintages, and the anomaly decile panels come from the original author's source but in a later rebuild that does not match the published descriptive statistics on any of the seven families. No result here is therefore eligible for an exact-replication label at any recovery rate. The consequence is asymmetric: the reconstruction can show that a published pattern reappears, and it can show that a pattern fails to extend, but it cannot attribute a cell-level discrepancy to the article rather than to the inputs. Where the audit falls short of published precision we classify the layer as partially recovered and say nothing stronger.

The second limitation is one of our own design. The registered materiality standard mixes two relative gates set at 10 percent with an absolute gate of 0.25 monthly percentage points, and on these test assets that absolute gate is equivalent to demanding a 52.6 percent reduction. The mixture makes the third gate roughly five times stricter than the other two, which we do not believe was intended, and it alone determines the headline classification. The threshold was frozen before any estimate existed and is applied as frozen; a future pre-registration should express the three gates on a common scale.

The third limitation is econometric, and it is the binding one for the whole paper rather than for the regime evidence alone. Cross-sectional pricing is fragile when betas carry a large share of sampling noise, and the calibrated simulation in Section 9.6 puts that share at 61.4 percent for rate betas even at 648 months, against 2.7 percent for market betas. Two consequences run through the results. Regime-length samples cannot resolve fitted-premium differences against the registered bound, which is why 44 of 70 equivalence tests are inconclusive. And the nominal Shanken interval for the rate price of risk attains its stated coverage at no simulated sample size, so the baseline t-ratio of -2.86 overstates the strength of the evidence behind it. Both statements are properties of the calibrated design rather than measurements of the real one, and the design flatters the estimator, so they bound this problem from the optimistic side. They are claims about this factor process and this cross-section, and we do not extend them further: another factor with a smaller noise share, or a different cross-section, need not behave this way. What generalises is the question rather than the answer. A study pricing a generated-regressor factor should establish the noise share of its own estimated exposures and the coverage of its own intervals rather than assume either.

A fourth limitation is that our registered identification standard does not test what its name suggests. The dispersion floor asks whether estimated exposures vary enough across assets for a second pass to run, and it is satisfied here by dispersion that is substantially noise. A future pre-registration should pair it with a reliability criterion on the exposures themselves.

The fifth limitation is interpretive. A priced covariance with a rate innovation would not by itself establish a narrow monetary transmission mechanism. Our regime evidence sharpens the point in an unwelcome direction: the exploratory break tests locate their breaks at dates that are not policy dates. We therefore separate pricing evidence from identification evidence throughout. The high-frequency policy-information decomposition, which was carried as an appendix design for exactly this purpose, has been withdrawn: the source was obtained and cannot answer the question, for the reasons in Appendix F.

11 Conclusion

This paper is a documented reconstruction, not a numerical replication, and the distinction governs what may be concluded from it. The article identifies its providers rather than its files, so we built the closest system obtainable from identifiable public inputs. On that system the published qualitative result recovers: short-rate innovations carry a negative and nominally significant price of risk in the cross section of anomaly portfolios, and adding the factor to the market model roughly halves cross-sectional pricing errors. Numerical recovery of the published cells is much weaker, particularly for pricing-error and fit statistics, and the audit in Appendix A reports it in full. Nothing here should be read as a check on the article's arithmetic; the inputs differ, and where a cell falls outside the published rounding we cannot say which of the two is responsible.

Implementing the article's own useless-factor bootstrap turns that caveat into a measurement. Its empirical p-values agree with all 45 published risk-price significance verdicts and with 27 of 29 each for the specification test and the cross-sectional fit, while landing inside the printed rounding far less often, because a five-thousand-replication p-value carries more Monte Carlo noise across most of its range than three printed decimals can resolve. The same holds for the risk-premium decomposition of the article's Table 5, whose 84 cells agree in sign without agreeing in digits. Inference reconstructs where arithmetic does not, which is the sharpest statement this design supports and a better one than a recovery rate alone.

The extensions are where the paper contributes, and both are negative. The short-rate factor does not meet the incremental-pricing standard we registered in advance, and every registered traded multi-factor comparator attains a smaller in-sample pricing error on the same seventy portfolios. The post-2013 estimates then fail every registered compatibility gate, reversing sign for five of seven anomaly families and deteriorating far beyond the registered bound. Two readings of that failure are ruled out and a third is not. It is not explained by the measured data-vintage contribution, which is smaller by three orders of magnitude, and the factor keeps clearing the registered exposure-dispersion gate after 2013. What we cannot rule out is imprecision: on 144 months the extension rate price carries a nominal Shanken t of -1.49, and four of the five sign reversals are of signs the baseline never established. The registered criterion fails; the underlying economic change is not thereby demonstrated.

A registered out-of-sample falsification puts the same fragility in forecasting terms. Refitting annually from a frozen 1999-12 endpoint and evaluating through 2025-12, the two-factor system attains a cross-sectional out-of-sample R^2 of 0.0039 against a historical-mean benchmark, against -0.1882 for a zero-return rule. The model is clearly better than assuming no risk premium and barely better than assuming the historical average, which is what a factor whose betas are mostly estimation error should be expected to deliver when it has to forecast rather than fit.

Across monetary regimes the evidence is asymmetric by construction. Exposures are demonstrably regime dependent under the pooled test, while the size of the change in fitted premia cannot be certified at feasible regime lengths in either direction. We report that asymmetry rather than resolving it, because the simulation shows the imprecision is a property of two-pass estimation on regime-length samples rather than a fact about the periods.

What survives is narrower than the original claim and, we think, more useful. The short-rate pricing pattern reconstructs qualitatively, and it is fragile as a general asset-pricing proposition: its incremental economic performance is threshold-sensitive, richer factor specifications fit the historical cross-section better in sample, its post-publication behaviour changes substantially, it improves on a historical-mean forecast by an economically negligible margin out of sample, estimated rate exposures are regime dependent, and the data are insufficient to estimate regime-specific pricing differences with the precision that strong economic conclusions would require.

We stop short of three stronger claims that the evidence does not carry. We do not say the factor ceased to be priced, because the post-2013 window is too short to separate a weakened relation from an imprecisely estimated one. We do not rank the short-rate model against the traded comparators, because in-sample pricing error on factor sets of different dimension is not a model-selection criterion. And we do not say the original result was wrong, because a reconstruction built from different inputs is not positioned to say so.

A Table-level replication audit

The table-level audit maps the article and supplement targets to replication statuses. It covers the article tables and the supplement appendix tables through Table A.14. Cell-level comparison was run against the statistics transcribed from Tables 3, 4, 5, 6 and A.1, which after collapsing rows that repeat a point estimate under two printed uncertainty measures give 215 unique published cells. All 215 were compared.

Table 11: Cell-level recovery by statistic. A cell counts as recovered when it agrees with the article to the precision the article prints, that is within half of the last printed increment.

Statistic	Cells	Recovered	Share	Median $\lvert\Delta\rvert$	Max $\lvert\Delta\rvert$
Market risk price	29	20	0.690	0.0033	0.0117
Rate risk price	16	3	0.188	0.0137	0.0901
Cross-sectional fit	29	3	0.103	0.0251	0.2863
Specification statistic	29	0	0.000	0.7458	7.4110
Constrained fit	5	1	0.200	0.0162	0.0492
Decile premium, market factor	21	9	0.429	0.0076	0.0277
Decile premium, rate factor	21	5	0.238	0.0116	0.1556
Decile pricing error	21	4	0.190	0.0142	0.0940
Decile average return	21	3	0.143	0.0137	0.0808
Comparator factor prices	15	2	0.133	0.0193	0.1350

Table 12: Replication layer classification. No layer is eligible for an exact-replication label, because no input is an exact article input.

Layer	Recovered	Classification
R1a short-rate innovations	per series	approximately reproduced under documented reconstruction
R1b first-pass betas	14 of 42	partially recovered under documented reconstruction
R1c risk prices	23 of 45	partially recovered under documented reconstruction
R1d pricing errors and fit	7 of 79	partially recovered under documented reconstruction
R1e comparator models	3 of 20	partially recovered under documented reconstruction

R1b is evidenced indirectly. The article tabulates no beta, so no cell compares a loading directly, but Table 5 tabulates beta times lambda for the extreme deciles. With the risk prices audited separately under R1c, a recovered premium is evidence about the loading behind it, and those 42 cells are classified here rather than left outside every layer. The average-return column of the same table is a property of the test assets rather than of an estimated layer, so it is compared cell by cell and belongs to no layer.

Three of the supplement's evaluation exercises are reconstructed, and what they have in common is that the supplement prints them. Its second fit measure replaces the variance in the denominator of equation (6) with the cross-sectional second moment, and all three of its published values are recovered: 0.92, 0.98 and 0.97 for the CAPM and the two short-rate models, against 0.9170, 0.9765 and 0.9726 here. Its covariance representation prices the cross-section on factor covariances rather than betas, and the supplement predicts the outcome, that the fit will equal the beta representation's. That is an identity rather than a coincidence, because the two designs span the same column space, and both routes are computed here and agree to within 3e-14 on the prices. The rate covariance price is -2.045 against a market price of -0.007, matching the supplement's description of an insignificant market price beside a strongly negative hedging one. Those prices are not comparable with the published ones in level: they differ by a factor near 102 on the rate price in both versions of the model, and the appendix does not state the units its factors carry, so the gap records a convention rather than a disagreement. The fit from the same table is a ratio of variances, carries no units, and reproduces the beta representation's own distance from its published value. Its stochastic-discount-factor form gives a Hansen-Jagannathan distance of 0.4127 for the funds-rate model and 0.4277 for the Treasury-bill model, against 0.4473 for the CAPM. The published distances are 0.419, 0.437 and 0.460, so none of the three lands inside the printed rounding, every one is low by between 0.006 and 0.013, and the ordering the supplement's report of a rejected CAPM and an unrejected ICAPM rests on is the same in both. The p-values beside them are not compared, because the supplement does not state the distribution they come from and this reconstruction generates no counterpart to it.

What remains unreconstructed divides in two, and the distinction matters more than the count. Some statistics are not specified by the source at all. The supplement's $\widehat{Q}_c$ statistic and its asymptotic p-values come from MATLAB code its footnote thanks Kan and Robotti for supplying; the distribution of the Hansen-Jagannathan distance under the null is not stated; the coefficient t-ratios of the last table come from a sequential procedure it cites rather than defines; and the covariance representation's standard errors come from a GMM covariance whose Shanken counterpart here is a different object. Implementing any of these would reconstruct the paper being cited rather than this article, so none is attempted and none is substituted. The rest are blocked by data: the augmented model needs six state-variable series this project does not hold, and two portfolio tables need sorts it does not have. Equal-weighted results are blocked at the current sources.

B Data access and reproducibility

Restricted article and supplement files remain local-only. The committed repository records metadata and source-access status, but it does not redistribute copyrighted source documents, prompt files, licensed data extracts, local catalogs, temporary artifacts, or machine-specific environment manifests. The source-only release gate reports zero critical restricted-path issues and one major missing-input issue.

Source-access requirements, release artifacts, and validation status are recorded outside the main argument in the repository documentation and generated release manifests.

Table 13: Original table groups and registered targets

Original evidence group	Registered targets	Status this pass
Article descriptive and pricing tables	TBL_001–TBL_009	Compared where a statistic-level target exists
Supplement interest-rate robustness	APP_TBL_A01–APP_TBL_A06	Partially compared
Supplement portfolio and model extensions	APP_TBL_A07–APP_TBL_A14	Not attempted this pass

Source: `research/table_target_manifest.csv`.

C Hypothesis registry

The confirmatory registry contains a replication target and hypotheses for incremental pricing content, post-publication compatibility, regime equivalence, and weak-factor strength. The registry distinguishes null, alternative, primary outcome, estimator, sample, test assets, economic direction or equivalence criterion, multiplicity family, and supported, unsupported, or inconclusive interpretation. Exploratory structural-break alignment is recorded separately from confirmatory claims.

D Table and figure architecture

Every result table and figure in this paper is generated from the frozen artifacts rather than transcribed. `scripts/build_manuscript_tables.py` writes the eight result tables and `scripts/build_manuscript_figures.py` the two figures; the manuscript includes the emitted files, and each generated row carries a comment naming the artifact it came from. A test regenerates every table and fails if the committed file differs, so a regenerated artifact cannot leave a stale number in the paper.

The registered architecture also anticipated figures for beta-return relations, rolling risk prices, and regime equivalence intervals. Those are not produced here. The equivalence intervals in particular would invite reading a per-portfolio contrast that Section 9 shows the sample cannot resolve, and we prefer to state that limitation in text than to draw it.

E Detailed out-of-sample design

The out-of-sample design freezes an initial training endpoint at 1999-12 and uses a predeclared annual refit schedule. Forecast records store model vintage, training dates, factor definition, asset universe, benchmark model, and loss function. Results are evidence about the stability of pricing errors, not a substitute for the baseline replication audit.

The design has now been run. The two-factor system is refitted annually from the frozen endpoint and evaluated on the seventy anomaly portfolios across 26 windows through 2025-12. The evaluation uses the vintage-consistent panel rather than a splice of the baseline and extension panels, because the window crosses the 2013-12 vintage boundary and a splice would let a data revision enter as an apparent forecasting result.

The model attains an out-of-sample R^2 of 0.0039 against the historical-mean benchmark, and the zero-return benchmark attains -0.1882 against the same reference. The reading is that the factor model beats a zero-return rule comfortably and beats the historical mean by a margin too small to be economically interesting, which is consistent with the in-sample picture in Section ef-sec:identification: a factor whose betas are mostly estimation error prices poorly where it has to forecast rather than fit.

The shipped table marks which models fall inside a fixed band of the lowest observed loss. That column is a descriptive screen and not the model confidence set of Hansen et al. (2011): it involves no resampling, no equal-predictive-ability statistic, and no coverage guarantee, so a model inside the band has not been shown to be equivalent to the leader. The selection rule travels in the table so the two cannot be conflated.

F Retired policy-information decomposition

This decomposition was carried as an appendix design on the condition, fixed in advance, that the monthly aggregation must pass factor-strength diagnostics before entering the pricing argument, because aggregating announcement windows to months can leave a factor that is zero in most of them. It is now withdrawn, and the condition that withdraws it is that one.

The registered source was obtained and examined rather than assumed unavailable. It covers 1990-02 onward, so it reaches 287 of the 504 baseline months and does not see the first eighteen years, which contain the Volcker disinflation and the largest short-rate movements in the sample. A decomposition estimated on what remains cannot be a check on the baseline pricing result. Under the primary identification its central-bank-information component is nonzero in 99 of 408 months. Set against a rate factor whose betas already carry a reliability ratio of 0.386 over 648 dense months, a component populated in a quarter of them would be identified from far less.

Both facts are properties of the source, not of an estimate, and both were read before any pricing regression used these data. The design is therefore not reported as tried and failed. It is withdrawn because the data cannot answer the question it was written to ask, and the AR residual keeps the label it has carried throughout: a rate innovation, not a policy shock.

G Additional diagnostics

Additional diagnostics include standardized exposure dispersion, rank checks, factor spanning, influence diagnostics, leave-one-family systems, boundary-shift checks, and multiple-testing adjustments. Raw beta dispersion, singular values, and condition numbers are descriptive diagnostics only. Diagnostic outputs remain pending until the relevant baseline and extension artifacts exist.

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